



Manish Agarwal

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Visa: H1B, i140 Approved

Professional Summary

With 16+ years of experience in designing and developing Java and Dot Net based web and mobile applications, specializing in **Angular, React, HTML, CSS, Java, Microservices Architecture, SQL/NoSQL** databases and containerization technologies (**Docker, Kubernetes**). Strong expertise in **micro front-end solutions** using Angular and other modern frontend frameworks. Proven leader in mentoring and leading teams, ensuring the delivery of high-performance, secure, and scalable solutions. Skilled in **CI/CD implementation on Azure DevOps and Jenkins** and deployment strategies, and in adopting software design patterns to meet evolving technical challenges.

Education

- **Master of Technology**, Data Science and Engineering | 2021 - 2023
Birla Institute of Technology and Science, Pilani, Rajasthan, India
- **Bachelor of Technology**, Computer Science | 2005 - 2009
Institute of Engineering & Technology, Alwar, Rajasthan, India

Training and Certifications

- Microsoft Certified Azure Fundamentals
- Microsoft Certified Data Fundamentals
- Microsoft Certified AI Fundamentals
- Successfully completed training on MongoDB provided by MongoDB University

Skills Set

Programming Languages	Java, Dot Net, Node JS, Python
Web Technologies and Frameworks	Spring Boot, Angular, React, Redux, NgRx, RxJS, Knockout JS, HTML 5, CSS 3, SASS, Object Oriented JavaScript, ECMAScript 6, Typescript, jQuery Mobile, Bootstrap, Unit Testing Frameworks (JUnit, QUnit, Jasmine, Karma)
APIs and Middleware	REST, Microservices, Web API, Spring JPA, Entity framework
Database	SQL Server, MongoDB
Tools and platforms	Microsoft Power BI, Browser Dev Tools, Lighthouse, Postman for API Testing, Insomnia, IntelliJ, Visual Studio Code, Mongo Compass, Jupyter Notebook, Jenkins, JIRA, Confluence, OpenShift, Datadog

Employment History

Organization	Position	Location	Duration
Nagarro	Staff Engineer	Jacksonville, FL	January 2015 - Present
Nagarro (India)	Lead Software Engineer	Gurugram, India	December 2010 – January 2015
Acidaes Solutions Pvt. Ltd.	Software Engineer	Noida, India	August 2009 – December 2010

Projects

Project Name	Finance & Real Estate
Client	Client is a US based Railroad Transportation Company.
Duration	July 2025 – Present
Description	The application has a group of several modules covering Finance and Real Estate applications from external Vendors. It provides them user friendly interface for various business transactions. It allows customers to request enquiries related to Real Estate, submit applications, track them, check for approvals, initiate contract and make payments. This application also has a modules allow users to submit their rental/purchase applications, create and manage invoices, initiate and manage projects, budget allocations and much more.
Technology Stack	Angular 15, Java 17, Microservices, Spring Boot/Spring Batch, Web API, Oracle DB, Jenkins (CI/CD), Git (Repo Management)
Responsibilities	• Serving as Team Lead, managing cross-functional teams, overseeing task assignments on Jira, and ensuring timely completion of user stories and deliverables. • Designed and developed new application features while ensuring the long-term maintainability of existing functionality, delivering high-quality, scalable code. • Architected and implemented key performance improvements by identifying and resolving bottlenecks on frontend and backend modules, resulting in significantly enhanced application performance, reduced latency, and improved throughput. • Achieved a major performance improvement after carefully refactoring and optimizing stored procedures. • Provided architectural guidance and mentorship to junior developers, ensuring the implementation of best coding practices and ensuring timely completion of user stories and deliverables.

Project Name	Price Selection and Maintenance
Client	Client is a US based Railroad Transportation Company.
Duration	August 2024 – June 2025
Description	Project is a web-based application that is primarily used by the Sales team to create contracts and prices, and publish them for further review by the MS teams and then MM team. App provides speed and simplification for price-related activities in revenue, accounting, sales and marketing, and marketing services.
Technology Stack	Angular 15, Java 17, Microservices, Spring Boot/Spring Batch, Web API, Mongo DB, Jenkins (CI/CD), Git (Repo Management), Datadog (Monitoring Tool), Jasmin, Karma, Junit, Postman (API Testing), Swagger
Responsibilities	• Developed and optimized new and existing microservices-based Java application, driving end-to-end enhancements in both the frontend (Angular) and backend (Java) architecture. • Implemented new features in the Angular application that improves user efficiency. • Architected and implemented key performance improvements by identifying and resolving bottlenecks in microservices and Spring Batch modules, resulting in significantly enhanced application performance, reduced latency, and improved throughput. • Utilized Datadog for proactive system monitoring and diagnostics, performed comprehensive root cause analysis on recurring issues (like memory leaks, long running Mongo DB queries among other issues).

- **Achieved a major performance improvement** after carefully refactoring and optimizing APIs taking hours to process and brought the **response time to 7-10 min.**
- **Developed unit test cases for both the frontend (Jasmine & Karma) and backend (JUnit) services**, increasing code reliability and test coverage.
- Responsible for testing APIs for requests and responses using Postman and Swagger.
- Led peer code reviews and guided the team in adhering to industry best practices, while managing the Git workflow, overseeing **Jenkins build pipelines**, and coordinating seamless production deployments to ensure efficient, reliable releases.

Project Name

Client

Duration

Description

Technology Stack

Responsibilities

Intermodal Asset Tracking

Client is a US based Railroad Transportation Company.

June 2023 – July 2024

Project is a web-based asset tracking application that involves locating the current position of all categories of railroad assets within the terminal. Device sends continuous location feeds to the server which the application displays on the client browser.

Angular 15, NgRx, MapBox, PrimeNG v15, React, Redux, Java 11, Microservices, Spring Boot, Spring JPA, Spring Batch, Kafka, MQTT, SQL, ClickHouse DB, MongoDB, Logstash, Jenkins, Jasmine, Karma, Junit, OpenShift, Git

- **Designed and developed new application features** while ensuring the long-term maintainability of existing functionality, delivering high-quality, and scalable code.
- **Architected and executed the migration of key features to Angular 15 & PrimeNG v15**, modernizing the frontend architecture with **micro-frontend design** principles and improving overall performance.
- **Integrated MapBox library** into Angular app that displays current location of drivers on the terminal, in addition to laying out terminal boundaries.
- **Developed React JS** module as part of the project which was used by Administrators to manage terminal activities.
- Also **served as Team Lead**, managing cross-functional teams, overseeing task assignments on Jira, sprint planning, and ensuring timely completion of user stories and deliverables.
- **Developed Microservices** that were being consumed by web applications to read driver coordinates that were streamed through **Kafka**.
- Collaborated with Business Partners in **finalizing the UX and UI designs**.
- Responsible for generating the build for all environment on **Jenkins** and administering version control, performing code reviews, and managing code integrations.
- Led troubleshooting and performance tuning to ensure a seamless user experience.
- **Developed and maintained unit and integration tests** for both frontend components (**Jasmin & Karma**) and backend services (**JUnit**).

Project Name

Client

Duration

Description

Technology Stack

Responsibilities

Contract Manager

Client is a US based sustainable resource management company.

November 2017 – April 2023

Project uses technology and people expertise to provide lower costs, drive efficiencies and reduce environmental impact for energy, water, waste and telecom usage. Project involved providing clients with insights into energy market along with recommendations, so they could make informed decisions. It helped clients develop an energy procurement strategy that aligned with the right product that meet their needs, their risk tolerances and which was timed with market buying opportunities.

Java 8, Web API, OOJS, JavaScript, jQuery, KnockoutJS, MS SQL, ETL, JPDL Workflow

- **Architected JSP based legacy system into modern, scalable and responsive web application** using Web APIs and Knockout JS.
- **Developed and automated JPDL workflows**, streamlining user task management and enhancing overall system productivity.
- **Optimized large-scale data system** being used via **SQL queries and Stored Procedures** allowing users to access millions of records without affecting their user experience.
- **Designed and implemented real-time notification systems using KnockoutJS**, allowing users to receive instant updates on task assignments and contract recordings, improving workflow efficiency.
- **Optimized various SSRS reports and ETLs** bringing the report generation time down from 16 hours to 10 minutes.
- **Provided architectural guidance and mentorship** to junior developers, ensuring the implementation of best coding practices and ensuring timely completion of user stories and deliverables
- Created and managed **CI/CD pipelines on Azure DevOps** platform along with managing code repositories on Azure.

Project Name

Client

Duration

Description

Technology Stack

Responsibilities

Mobile Rail Tool

Client is a US based Railroad Transportation Company.

December 2015 – April 2017

MRT is a mobile browser-based application that works in online as well as offline mode. This application is being used by the conductors on the rail cars to perform a predefined work during their route from origin to the destination. Earlier they were doing paperwork to manage that task. Now they can perform it through mobile. This application has an admin role which is used to generate a work order and assign that to a conductor. As soon as the work order is created, corresponding conductor is informed on the call to download the work order on his device and start performing his task. The tasks involve pickup, place, pull and setting out cars at different stations, customers and/or junctions. The person using this application can not only perform the designated tasks but also add work on the fly and report it.

Java, Rest API, OOJS, jQuery, Angular JS, Oracle, IBM Mainframe

- Involved in initial POCs for the feasibility of the concept on tablets and desktops.
- Responsible for analysis, estimation, design and development of various components of the project
- Involved in Code Reviews of modules developed by team.

- Involved in work allocation, progress monitoring, and mentoring of team members.
- Responsible for code optimization and taking code reusability initiatives.
- Involved in fixing/tracking sprint bugs till closure.
- Followed up and coordinated with offshore team while on onsite assignment to US.

Project Name

Client

Duration

Description

Technology Stack

Responsibilities

Financial Planning Application (For Students)

Client is a US largest organization dedicated to giving young people the knowledge and skills they need to own their economic success, plan for their futures, and make smart academic and economic choices.

September 2013 – May 2014

Project is divided into two applications: Simulation and Content Management System. Simulation is a Single Page Web application running on tablets, iPads, desktops, laptops & notebooks. It is being used by students studying in grades 7 to 12. While CMS is an MVC based web application that runs on desktops and provides complete management of content like sponsor logos, videos/presentation, pictures, text descriptions and other master data.

ASP .Net MVC, jQuery, Backbone JS, HTML 5, CSS 3, MS SQL

- Involved in gathering and understanding client requirements.
- Developing application architecture for mobile/web-based applications.
- Formed JavaScript Coding Guidelines and Best Practices for team.
- Responsible for technical delivery from development standpoint.
- Created knowledge transfer documents for smooth transition.

Project Name

Client

Duration

Description

Technology Stack

Responsibilities

Knowledge Navigator

Client is a German based luxury vehicle, motorcycle, and engine manufacturing company. It is one of the best-selling luxury automakers in the world. The project was a part of their innovation systems.

February 2012 – August 2013

Project involved POC and then development of a system for an effective and profitable usage of all the knowledge stored in the knowledge database of the client. The system was based on the principle "Knowledge should be accessible to anyone, anywhere and anytime". The system includes development of 3D view of knowledge graph, with voice as input commands to provide more flexibility to user and an interactive way to learn about knowledge. The system was developed to support Leap Motion, Pixel Sense (Microsoft Surface) and Voice Commands.

Dot Net C#, XAML, Lead Motion SDK, Microsoft Surface SDK, MS SQL

- Involved in gathering and understanding client requirements.
- Involved in initial POCs for the feasibility of the concept on Microsoft Surface and Desktops using Leap Motion
- Involved in gathering and understanding client requirements.
- Responsible for analysis, estimation, design and development of various components of the project
- Involved in work allocation, progress monitoring, and mentoring of team members.
- Responsible for code and performance optimization and taking code reusability initiatives.
- Involved in fixing/tracking bugs till closure.
- Responsible for technical delivery from development standpoint